

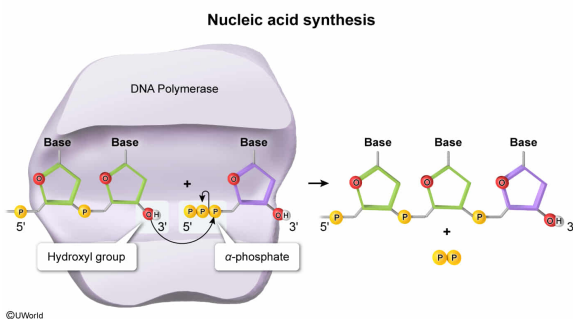
Which of the following correctly describes what happens when DNA polymerase adds a deoxyribonucleotide triphosphate (dNTP) to a growing DNA strand?

- ☐ A. The hydroxyl group at the 5' end of the growing strand attacks the 3' α -phosphate of the dNTP. [12%]
- ☐ B. The phosphate group at the 5' end of the growing strand attacks the 3' hydroxyl group of the dNTP. [20%]
- ✓ ☒ C. The hydroxyl group at the 3' end of the growing strand attacks the 5' α -phosphate of the dNTP. [55%]
- ☐ D. The phosphate group at the 3' end of the growing strand attacks the 5' hydroxyl group of the dNTP. [11%]

Correct

55% answered correctly

Explanation:



Nucleotides are biological molecules composed of the following:

- One **pentose sugar** (ie, ribose in RNA or deoxyribose in DNA)
- One to three phosphates
- One of the five **nitrogenous bases**: adenine, cytosine, guanine, thymine (DNA only), or uracil (RNA only)

The nitrogenous base is attached to carbon 1 of the sugar, all phosphates are attached to carbon 5 (ie, the 5' end), and a hydroxyl group (OH) is attached to carbon 3 (ie, the 3' end).

Nucleotides can combine to form long chains known as polynucleotides or nucleic acids (ie, RNA or DNA), as catalyzed by **RNA polymerase** or **DNA polymerase**. In biological systems, lengthening of these chains occurs from the **5' end to the 3' end** (ie, **new nucleotides are added to the 3' end of the chain**). During this process, the **3' hydroxyl group** of the growing chain acts as a **nucleophile** to **attack the 5' α -phosphate** of the **incoming nucleotide**.

(Choice A) The hydroxyl group of the growing chain is found at the 3' end, not the 5' end.

(Choice B) The phosphate group at the 5' end of the growing strand is not involved in the

reaction because new dNTPs are added to the 3' end of the growing strand.

(Choice D) The 3' end of the growing strand contains a hydroxyl group, not a phosphate group.

Educational objective:

Nucleotides consist of a pentose sugar linked to a nitrogenous base at carbon 1, one or more phosphates linked to the 5' end, and a hydroxyl group linked to the 3' end. Nucleic acids grow when the 3' hydroxyl group of a chain acts as a nucleophile to attack the 5' α -phosphate of an incoming nucleotide, as catalyzed by RNA polymerase or DNA polymerase.

Time spent:0 sec

Subject:
Biochemistry

QID:403098

System:
Carbohydrates, Nucleotides, and Lipids